PIC MICROCONTROLLER-BASED ELECTRONIC LOCK



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An electronic lock allows activation of an electric appliance only on entering the correct password. Here we present such an electronic locking system in which a PIC16F877A microcontroller plays the role of the processing unit. The MCU is interfaced with a 4×4 matrix keypad and a 16×2 LCD to form the user interface. Using this circuit, you can make any electrical appliance password-protected. It can also be used as an electronic door lock by interfacing the output of the circuit with an electrically actuated door lock. The system turns on the appliance on entering a four-digit password set by the user.

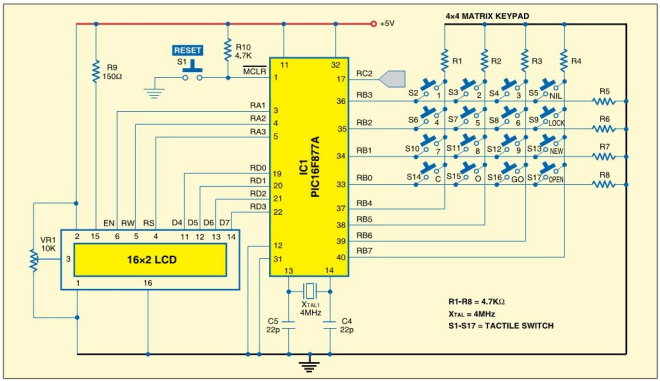


Fig. 1: Circuit of PIC microcontroller-based electronic lock

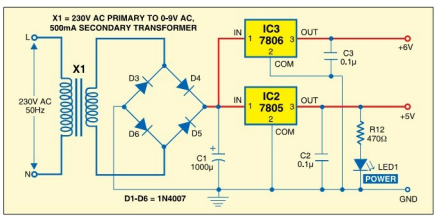


Fig. 2: Power supply circuit

Circuit description

Fig. 1 shows the circuit of the PIC microcontroller-based electronic lock. It can be divided into five sections: input (4×4 matrix keypad), processing unit (PIC16F877A MCU), appliance controller (relay driver), display (16×2 LCD), and power supply.

PIC16F877A MCU. The PIC-16F877A is an 8-bit microcontroller based on reduced instruction set computer (RISC) architecture. It has 8k×14-